05 LECTURE HEALTH CARE

MT 1, Second Semester LECTURE 05 - Sir Micheal Gu, RMT

COMMUNITY WATER, WASTE MANAGEMENT, AND SEWAGE DISPOSAL

TOPIC OVERVIEW

LEARNING OBJECTIVES

1. Understand the **relationship between water, waste, and public health**
2. Identify **sources of water and waste contamination**
3. Learn **water treatment and waste disposal methods**
4. Explore **sustainable waste management practices**
5. Recognize **chemical safety protocols** in waste and water management

I. INTRODUCTION TO ENVIRONMENTAL HEALTH

A. ENVIRONMENTAL HEALTH

- Environmental health **examines external factors that affect human well-being**
- It studies the factors that directly affect our **public safety**, **economic development**, and **sustainability** in our community.
- **Factors/Key components:**
  - **Water quality (Clean Water)**
    - Ensuring clean drinking water and wastewater management
  - **Air quality (Clean Air)**
    - Pollution control, emission reduction
  - **Waste management**
    - Proper disposal, recycling, and composting
  - **Sanitation and hygiene**
    - toilets , handwashing, disease prevention
- If these four are neglected, there will be poor outcomes

B. IMPACT OF POOR HEALTH AND WASTE MANAGEMENT

- **Health impacts**
  - Contaminated water causes diarrheal diseases, cholera, and typhoid
  - Exposure to waste increases respiratory infections and skin diseases
- **Environmental impacts**
  - Polluted rivers and lakes lead to the destruction of marine life
- **Economic and social impacts**
  - Poor sanitation increases healthcare costs
  - Contaminated environments reduce tourism and economic growth

- **Poor outcomes** can lead to *disease outbreaks*, *water pollution*, and *environmental damage*.
- **Improper waste disposal** has serious consequences.

- **Water pollution** increases the risk of diseases, which in turn creates a heavier burden on healthcare costs, whether for individuals (personal) or the government.
- Pollution damages ecosystems, which affects **livelihoods** and **tourism**.
- In the case of the Philippines, industries such as *island hopping* and *whale shark watching* (Oslob) can be negatively impacted by water pollution.
- Proper waste management is essential for maintaining a healthy and well-managed community.

I. WATER IN THE COMMUNITY (SOURCES, TREATMENT, QUALITY)

A. IMPORTANCE OF WATER

- Content:
  - Water is the most critical natural resource, yet billions don't have access to it.
  - The human body is 70% water - essential for survival
- Uses of water:
  - **Drinking & Cooking**: Safe hydration and food preparation
  - **Sanitation & Hygiene**: Bathing, washing, and cleaning, especially on the laboratory machine, by cleaning it to avoid possible cross-contamination.
  - **Agriculture & Industry**: Irrigation, food production, and manufacturing
- **Global Water Crisis**:
  - **2.2 billion people** lack access to safe drinking water.
  - Climate change and pollution reduce water availability

- Water is our most critical natural resource, yet millions of people do not have access to safe drinking water.
- It is essential for survival and hygiene, and also a necessary component in food production and many industries.
- Even in the laboratory, we are highly dependent on water to clean our machines after every sample run. Without properly cleaning our machines, test results can be affected.
- However, because of human activities, our water supply is rapidly decreasing due to pollution, overuse, and even climate change.

B. SOURCES OF WATER IN THE COMMUNITY

- Water comes from surface water sources, groundwater sources or rainwater harvesting

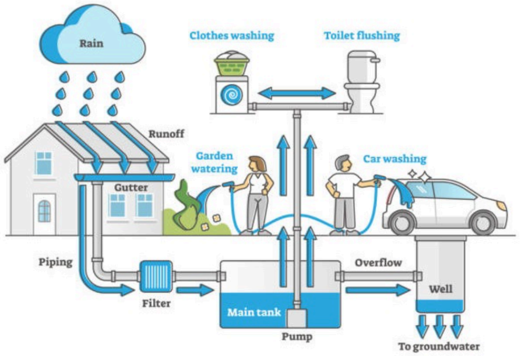
Surface Water sources

- Surface water sources come from the surface of the Earth (rivers, lakes, reservoirs, streams).
- It provides water for millions of people.
- It is the most common source of drinking water in urban areas.
- Examples:
  - **Rivers**: Nile, Amazon, Ganges
  - **Lakes**: Victoria, Michigan, Baikal
  - **Reservoirs**: Man-made lakes storing water for human use
- **Pros**:
  - Easy to access
  - Can store large amounts of water
- **Cons**
  - Limitation: Highly vulnerable to pollution (industrial waste, sewage, and chemicals)
  - Requires proper treatment before it is considered safe for human consumption

Groundwater sources

- Water is stored beneath the earth's surface in aquifers, wells, and springs
- Naturally filtered through soil and rock layers.
- **Examples:**
  - Wells & Boreholes: Access underground water for drinking and irrigation.
  - Springs: Natural outflow of groundwater.
- **Pros**:
  - Generally cleaner due to natural filtration. (Cleaner: Ground Water > Surface Water)
  - Less exposed to pollution.
- **Cons**:
  - Over-extraction can deplete groundwater levels.
  - Risk of contamination from septic tanks, industrial chemicals, and pesticides
    - Overextraction and contamination from chemicals can lead to groundwater becoming unsafe

Rainwater Harvesting



- This is a simple and effective way to conserve water.
- The process of collecting and storing rainwater for household, agricultural, or industrial use.
- Still considered a surface water source because it is highly exposed to pollution and requires proper storage
- Methods of Rainwater Harvesting:
    - **Rooftop Catchment Systems**: Water collected from rooftops and stored in tanks or underground reservoirs.
    - **Surface Runoff Harvesting:** Rainwater collected from the ground into ponds, dams, or underground reservoirs.
    - **Rain Barrels & Storage Tanks**: Small-scale collection for gardening and household use.

- It allows households and communities to store water for later use.
- However, technically, this is still a (artificial) *surface water source*, which means it is highly exposed to pollution.
- **Proper storage** is essential to prevent it from becoming a breeding ground for mosquitoes (reduce the spread of dengue and malaria)..
- **Filtration** is necessary before consuming rainwater, especially in today's environment, where **acid rain** is becoming more common.

- **Benefits**:
    - **Reduces dependence** on municipal/city water supply.
        - *Cebu City: MCWD (reduced dependence = low water bill)*
    - It provides a **free and sustainable water source**, especially in our tropical country where rainfall can be irregular.
        - *Collected rainwater can be used for washing cars, cleaning homes, and watering plants*
    - Helps **recharge groundwater levels**.
    - **Reduces stormwater runoff** that causes flooding through the community effort of rainwater collection
- **Challenges & Limitations**:
    - **Requires proper filtration** to prevent contamination and pollution.
    - Not effective in **low rainfall areas.**
    - High **installation costs** for large systems

WATER TREATMENT METHODS

- Why Water Treatment is Necessary:
    - Natural water sources often contain bacteria, viruses, sediments, and chemical pollutants.
    - Treatment ensures water is safe for drinking and household use.

TREATMENT METHODS

1. **Boiling**

- **Boiling** is the simplest and an effective method to kill bacteria and viruses.

Simple Boiling

- In ***simple boiling***, water is heated to **100°C** (212°F) to kill microorganisms.
- Simple Boiling Disadvantage: may not remove any *chemical pollutants*

Pasteurization

- Another form of boiling is ***pasteurization***
    - It is a form of boiling; however, the temperature used is lower and controlled
    - It is heated at **62.5°C (or 62-63°C)**
        - Why?
            - To not break down protein molecules inside the milk (for safe consumption)
            - To ensure it only removes harmful bacteria like *E. coli* but retains good bacteria like *Lactobacillus*.
    - The principle behind pasteurization is that most disease-causing microorganisms are destroyed at temperatures lower than boiling

    - Therefore, water does not need to reach 100°C to become microbiologically safer
        - For comparison, when proteins are exposed to high heat (like in milk during cheesemaking), they denature and coagulate
        - Pasteurization avoids extremely high temperatures to prevent major chemical changes

Effectiveness:
    - Eliminates **bacteria, viruses, and parasites**
    - **Does NOT remove chemical pollutants** like lead, mercury, or pesticides

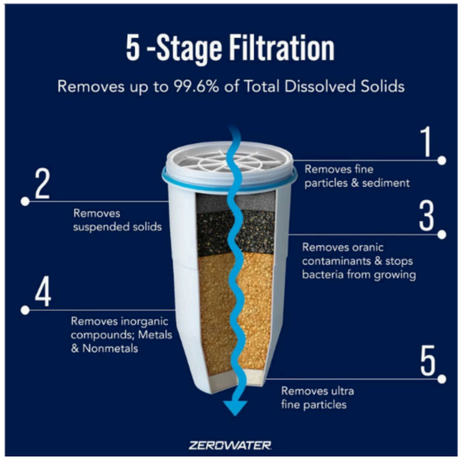
Distillation

- Another form of boiling is **distillation**, which can remove chemical pollutants (though not all)
- During distillation, white residues may appear in the flask, indicating remaining chemical contaminants
- Unlike simple boiling, this is not typically seen in common household

Tyndallization

- To have sterile water, there is one thing you can do with boiling
- Sterile water can be obtained by **boiling water 3 times**, a process called **Tyndallization**
- It is not a common or recommended method for purifying drinking water
- This method is also rarely used in modern medicine today, as more effective techniques (autoclaves) are now available
    - First Cycle: Boil it and let it cool down (initially kill most of the organisms)
    - Second Cycle: Boil again and cool down
    - Third Cycle: Boil one last time and cool down (to ensure every microorganism is killed)

2. Filtration



- Removing **physical impurities** by means of a **filter**
- It is sufficient in that function, but fails to eliminate microorganisms and certain chemicals (only chemicals that have large molecular sizes)
- **Filtration** is commonly used *together with Simple Boiling* in many households to ensure that water is safer for consumption.
- Filtration alone can be insufficient.
- **Process:**
    - Water is passed through porous materials (sand, ceramic, activated carbon) to remove contaminants.
- **Effectiveness:**
    - Removes dirt, debris, and some bacteria.
    - Activated carbon filters remove chlorine and organic compounds.
    - Does NOT remove viruses or dissolved chemicals unless specialized filters are used

Method	Removes Bacteria	Removes Viruses	Removes Chemicals	Best for?
Boiling	Yes	Yes	No	HouseHold, Emergency Use
Filtration	Some	No	Some	Continues Clean Supply

Chlorination and UV treatment

Chlorination

- **Process**:
    - Adding chlorine to water to kill bacteria, viruses, and parasites
- **Effectiveness**:
    - Used in city-wide water systems worldwide because of how cost-effective and efficient.
- **Limitations**:
    - Can produce chemical by-products that may produce a taste

UV Treatment

- **Process**:
    - Exposing water to ultraviolet light disinfects it.
- **Effectiveness**:
    - It is highly effective in killing bacteria, viruses, and protozoa.
- **Limitations**:
    - Requires electricity, so it is not always feasible in rural areas or

places where electricity is not stable

Large-Scale Water Treatment Plants

- Tycalayy uses in urban cities and uses multiple steps to ensure safe drinking water
- It starts from Coagulation to filtration to disinfection.
- Each stage ensures that each step removes bacteria and contaminants before household distribution

Stages of Water Treatment:

1. **Coagulation & Flocculation** - Chemicals added to bind dirt particles.
2. **Sedimentation** - Heavy particles settle to the bottom.
3. **Filtration** - Water passes through sand and charcoal filters
4. **Disinfection**– Chlorine or ozone kills remaining bacteria.
5. **Storage & Distribution**– Clean water is stored and sent to homes

II. WATER CONTAINMINATION AND SAFE HANDLING

A. WATER CONTAMINANTS

Physical and Chemical

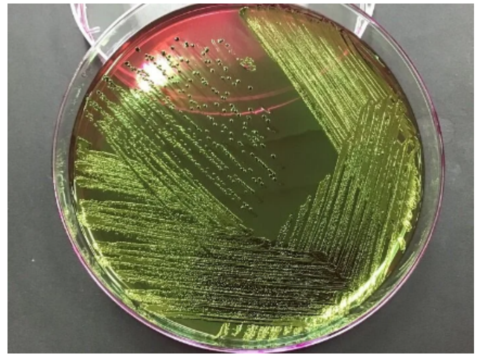
Physical Contaminants

- **Sediments, soil, plastic debris** – Common in surface water.
- **Cloudy water (Turbidity)** – Affects appearance and quality
- **Example**: Plastics or any physical trash human throws away

Chemical Contaminants

- **Heavy Metals (Lead, Mercury, Arsenic)**– Cause poisoning and organ damage.
- **Pesticides & Fertilizers** – Agricultural runoff contaminates drinking water.
- **Industrial Waste (Oil, Chemicals, Pharmaceuticals)** – Hazardous Pollutants.
- Example: Industrial waste from the factory that is dumped in the river. Fertilizer or even cooking oils (Please don't throw it in the sink).

Microbial



- **Bacteria**
    - **E. coli, Salmonella, Vibrio cholerae** – Cause severe gastrointestinal diseases.
    - **Salmonella typhi:** – Causing typhoid fever and Salmonella
    - **Vibrio cholerae** - Causes cholera
    - **Shigella dysenteriae** - causes dysenteries
- **Viruses**
    - **Hepatitis A, Norovirus, Rotavirus** – Cause stomach flu and liver infections.
- **Protozoa**
    - **Giardia, Cryptosporidium** – Found in untreated water sources and are resistant to chlorine(Not recommended to drink water directly from the tap)
        - If you have no choice to drink tap water, you need to boil it first

B. WHO AND NATIONAL WATER QUALITY STANDARDS

- A set of standards to ensure the water is safe from harmful bacteria, chemicals, and excessive mineralization
- It is set by our government to ensure that our water supply is protected

WHAT IS MINERALIZATION?

- Mineralization is the process of converting organic compounds into inorganic, mineral forms through microbial decomposition (soil science) or the deposition of inorganic materials into organic matrices (biology/geology)

- **World Health Organization (WHO) Standards**
    - Drinking water should have **zero coliform bacteria**.
        - Presence of coliform bacteria may indicate a **contamination of feces,** both human and animal
    - Chemical contamination must not go above a certain threshold level.
        - Example: The normal level of arsenic contamination of water to be safe according to WHO Standards are generally **below 10 g/L**
    - Maximum contaminant levels for heavy metals and chemicals
    - pH should be between 6.5 and 8.5 for safe consumption because at that range it is safe for consumption
        - Below 6.5 = chemical contamination of acidic nature
        - Above 8.5 = chemical contamination of basic or

alkaline in nature, or it also suggests a severe microbial contamination because when bacteria come into contact with liquid it creates a basic environment.

- **National Standards:**
    - Countries have their own water safety regulations.
    - Regular water testing ensures compliance with health standards.

C. SAFE HANDLING

- **Common Tests:**
    - **Coliform Bacteria Test** - Detects fecal contamination.
    - **Turbidity Test** – Measures water clarity.
    - **Pathogen-Specific Tests** – Identifies cholera, typhoid, and other harmful microorganisms Testing for Microbial Contamination.

Testing for Microbial Contamination

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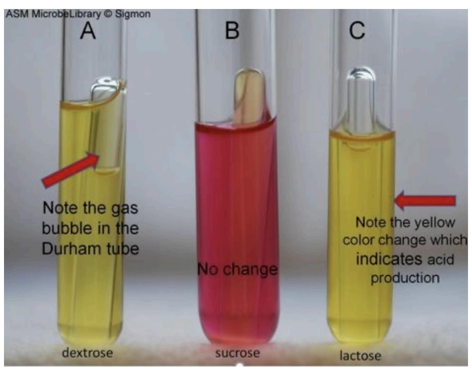
1. **Coliform Bacteria Test**



INTERPRETATION OF RESULTS

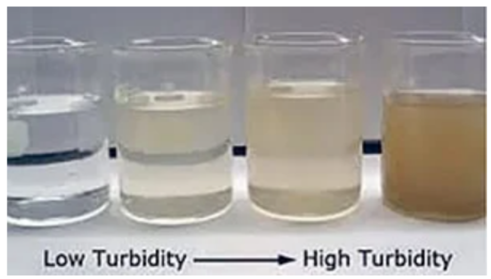
Test Tube A: Presumptive POSITIVE for coliform bacteria — lactose fermentation occurred, shown by **color change (acid production)** and **gas bubble in the Durham tube** (not safe)

Test Tube B: NEGATIVE for coliform bacteria — no color change and **no gas bubble**, indicating **no lactose fermentation** (safe)

Test Tube C: NEGATIVE (acid only) — color changed (acid production) but **no gas bubble** in the Durham tube, so it is **not counted as presumptive positive for coliforms**; proceed/retest if required by the lab protocol. (Not safe)

- **Note**: To ensure a safe water sample is considered safe, there must be **no change in color** and **no bubble formation.**

2. **Turbidity Test**



- The simplest test you can do for microbial contamination is turbidity test
- Pour water into a clear beaker to test for turbidity. If clear, it is presumably safe to drink

3. **Pathogen-Specific Tests**

- Basically test kit to test directly for E.coli, Shigella dysenteriae, and for those specific bacteria that cause harm to our health
- But It's Expensive :(
- **Common Chemical Tests:**
    - **Lead and arsenic testing** – Detects heavy metals.
        - Example: Microchip industry manufacturing
    - **pH analysis** – Ensures safe acidity levels.
    - **Nitrate and nitrite testing** – Prevents contamination from fertilizers. Safe Water Collection & Transport
        - This is done in an agricultural area because of possible fertilizer contamination.

Safe Water Collection & Transport

- **Best Practices:**
    - **Use sealed, clean containers for water storage.**
    - Avoid direct hand contact with drinking water.
    - Use covered buckets or jerry cans to prevent contamination Safe Water Storage

Safe Water Storage

- **Storage Guidelines:**
    - Keep water in cool, dark places to prevent bacterial growth.
    - Use airtight lids to prevent contamination.
    - Clean storage containers regularly with mild disinfectants. Preventing Contamination

Preventing Contamination

- **Key Steps:**
    - Separate clean and dirty water containers.
    - Do not dip hands or dirty utensils into stored drinking water.
    - Disinfect storage tanks and pipelines regularly

II. COMMUNITY WASTE AND SEWAGE DISPOSAL

A. TYPES OF WASTE

Main Types

- **Organic Waste:**
  - Includes food scraps, garden waste, and agricultural residues.
  - Can be composted into fertilizer.
- **Recyclable Waste:**
  - Includes paper, plastic, glass, and metals.
  - Can be reused or processed into new products.
- **Hazardous Waste:**
  - Includes batteries, medical waste, chemicals, and toxic materials.
  - Requires special disposal methods to prevent environmental harm.
- **Industrial Waste**:
  - Byproducts from factories, power plants, and refineries.
  - Can include chemicals, heavy metals, and toxic sludge

Solid vs. Liquid Waste

- **Solid Waste:**
  - Includes household garbage, plastics, metals, paper, and construction debris.
  - Can be recycled, composted, incinerated, or landfilled.
- **Liquid Waste:**
  - Includes sewage, industrial wastewater, chemicals, and oil spills.
  - Requires treatment before disposal into water bodies

B. CONSEQUENCES OF IMPROPER WASTE DISPOSAL

- **Water Contamination:**
  - Pollutants seep into groundwater, rivers, and lakes.
- **Soil Pollution:**
  - Toxic waste makes soil infertile and harmful for agriculture.
- **Air Pollution:**
  - Burning waste releases toxic gases and greenhouse emissions.
- **Health Risks:**
  - Waste attracts disease-carrying pests (rats, flies, mosquitoes).
  - Increases the risk of respiratory infections, food poisoning, and waterborne diseases.

C. METHODS OF SOLID WASTE DISPOSAL

- **Landfilling:**
  - Most common method, but causes **groundwater contamination**.
- **Incineration:**
  - Burning waste to reduce volume, but **releases air pollutants**.
- **Composting:**
  - Converts organic waste into fertilizer for farming.
- **Recycling:**
  - Reduces waste by reusing materials like glass, paper, and plastic

D. WHAT IS SEWAGE?

- It is called liquid waste (a mixture of solid and liquid waste)
- Waste water from homes, businesses, and industries.
- **Components:**
  - Human waste (feces, urine).
  - Soaps, detergents, food scraps.
  - Industrial chemicals, grease, and heavy metals.
- **Health Risks:**
  - Can carry bacteria, viruses, and parasites.
  - Causes waterborne diseases if untreated

E. SEPTIC TANKS AND SEWAGE PLANTS

- **Septic Tanks:**
  - Small-scale wastewater treatment for rural areas
  - Requires regular emptying and maintenance.
- **Sewage Treatment Plants:**
  - Large-scale systems that process municipal wastewater.
  - Uses biological and chemical treatments to remove contaminants

F. IMPACT OF POOR SEWAGE MANAGEMENT

- **Water Contamination:**
  - Wastewater seeps into rivers and groundwater.
- **Disease Spread:**
  - Increases risk of cholera, hepatitis, and dysentery.
- **Environmental Damage:**
  - Kills fish and marine life due to oxygen depletion

G. WASTEWATER TREATMENT STEPS

1. **Screening:** Removes large objects (plastic, sticks, cloth).
2. **Primary Treatment:** Solids settle, grease and oil are skimmed.
3. **Secondary Treatment:** Bacteria break down organic matter.
4. **Tertiary Treatment:** Advanced filtration removes remaining contaminants.
5. **Disinfection: Chlorination** or **UV light:** kills harmful bacteria.

V. STREAM POLLUTION AND URBAN RUNOFF

A. WHAT IS STREAM POLLUTION

- Stream Pollution occurs **when harmful substances enter rivers, lakes and**

streams. Making the water unsafe for drinking and damaging aquatic life.

- How to prevent stream pollution by enforcing strict regulations and promoting sustainable, safe practices

- **Sources of Stream Pollution:**
  - **Industrial Waste:** Factories discharge chemicals and heavy metals into rivers.
  - **Sewage Discharge:** Untreated wastewater enters streams, increasing bacteria and nutrient pollution.
  - **Agricultural Runoff:** Fertilizers, pesticides, and livestock waste contaminate water sources.
  - **Plastic Pollution:** Plastic debris clogs waterways and harms fish and birds.

B. CONSEQUENCES OF STREAM POLLUTION

- **Unsafe drinking water**
  - Requires expensive treatment
- **Loss of biodiversity**
  - Fish and aquatic life die due to pollution.
- **Dead Zones**
  - Excess nutrients cause algal blooms that deplete oxygen.

C. URBAN RUNOFF AND ITS CONSEQUENCES

- It is a form of liquid pollution that is exclusively found in the city (asphalted roads and motor vehicles)
- Rainwater that flows over **streets, sidewalks, and rooftops** collects pollutants before draining into waterways.
- Common Pollutants in Urban Runoff:
  - Oil and grease from vehicles
  - Heavy metals from roads and construction
  - Pesticides and fertilizers from lawns
  - Litter, plastic debris, and pet wastes

- **impacts**:
  - Overwhelms drainage systems, increasing flooding risks.
  - Introduces toxins and bacteria into rivers.
  - Reduces water quality and affects aquatic life

D. HOW AGRICULTURAL WASTE POLLUTES WATER

- **Common Agricultural Pollutants:**
  - **Pesticides & Herbicides**- Harmful chemicals wash into rivers, contaminating drinking water.
  - **Fertilizers (Nitrates & Phosphates)**-Excess nutrients cause algal blooms that deplete oxygen.
  - **Livestock Waste**- Animal feces introduce E. coli and other bacteria into water.
- **Impact on Ecosystems and Human Health**
  - **Dead Zones:** Excess fertilizers lead to low oxygen levels, killing fish.
  - **Drinking Water Contamination:** Nitrates in water cause blue baby syndrome in infants.
  - **Spread of Waterborne Diseases:** Animal waste introduces harmful bacteria into rivers

E. STRATEGIES TO PREVENT WATER CONTAMINATION

- **Improve Wastewater Treatment:**
  - Upgrade sewage plants to remove harmful chemicals.
  - Promote community sanitation projects.
- **Reduce chemical use in farming:**
  - Use organic pesticides and fertilizers.
  - Practice crop rotation and natural pest control.
- **Implement Green Infrastructure:**
  - Create vegetative buffer zones near rivers.
  - Install permeable pavements to reduce runoff
- **Have other methods of sheltering animals**
  - To prevent the spread of feces from animal droppings

V. SOLID WASTE MANAGEMENT AND RECYLING

A. WASTE SERGREGATION AND THE 3Rs

- **Reduce**
  - Minimize waste by using eco-friendly alternatives.
  - Example: Use reusable shopping bags instead of plastic.
- **Reuse**
  - Repurpose items instead of throwing them away.
  - Example: Glass bottles can be cleaned and reused for storage.
- **Recycle**
  - Convert old materials into new products.
  - Example: Plastic bottles can be recycled into textile fibers

B. COLLECTION, STORAGE AND TRANSPORT OF WASTE

- **Household Collection:**
  - Waste bins for biodegradable, recyclable, and non-recyclable waste.
- **Municipal Waste Collection:**
  - Garbage trucks collect and transport waste to disposal sites.
- **Industrial Waste Transport:**

- Factories must separate hazardous waste for proper treatment

VI. CHEMICAL SAFETY IN WASTE AND WATER MANAGEMENT

A. HANDLING AND STORAGE OF HAZARDOUS CHEMICALS

- **Why Proper Handling is Important**:
    - Prevents chemical spills and contamination of water sources.
    - Reduces exposure to toxic and corrosive substances.

- **Safety Guidelines:**
    - **Label hazardous materials** properly (clear warnings and hazard signs).
    - Store chemicals in secure, ventilated areas, away from food and drinking water.
    - Use protective equipment (gloves, masks, goggles) when handling dangerous chemicals
    - The most important label of chemicals is the **date created**
    - Bare minimum of labeling:
        - Name
        - Concentration
        - Date created
        - Who created

B. RISK OF CHEMICAL CONTAMINATION IN WATER AND WASTE

- **Sources of Chemical Contamination:**
    - Industrial waste is dumped into rivers and lakes.
    - Improper disposal of household chemicals (bleach, paints, batteries).
    - Agricultural runo (pesticides, herbicides, fertilizers).

- **Health Risks**:
    - Heavy metals (Lead, Mercury, Arsenic): Cause nerve damage and poisoning.
    - Pesticides & Herbicides: Linked to cancer, hormone disruption, and birth defects.
    - Chemical residues in water: Lead to long-term chronic illnesses

C. SAFE DISPOSAL OF CHEMICALS AND REGULATION

- **Government Regulations**:
    - Strict guidelines for hazardous waste disposal (Environmental Protection Agencies).
    - Industries required to treat chemical waste before disposal.

- **Safe Disposal Methods**:
    - Specialized hazardous waste facilities for industrial chemicals.
    - Community collection centers for household chemicals.
    - Promoting biodegradable & eco-friendly alternatives

D. THE ROLE OF GOVERNMENT POLICIES IN WASTE MANAGEMENT

- **Why Government Policies Matter:**
    - Regulate **industrial waste disposal.**
    - Prevent **illegal dumping** and fine violators.
    - Encourage **public awareness and environmental education.**

- **Key Policies & Regulations:**
    - Clean Water Act (USA & other countries)–Protects water from Pollution.
    - **Extended Producer Responsibility (EPR) -** Companies must manage waste from their products
    - **Plastic bans & Waste Reduction Laws** - Reduce single-use plastics

VII. SUMMARY

- Proper waste and water management prevent pollution and disease
- Sustainable practices like recycling and composting reduce landfill waste
- Government policies play a key role in regulating waste disposal
- Community participation is essential for effective waste management
- Challenges in waste and water management
    - Lack of awareness and public participation
    - Limited infrastructure for waste disposal in rural areas
    - High costs of sustainable waste management
- Future of waste and water management
    - Advancements in waste-to-energy technologies
    - Expansion of recycling and biodegradable packaging
    - More government policies to combat pollution
- Call to action - what you can do
    - Reduce waste by avoiding single-use plastics
    - Recycle and compost whenever possible
    - Support policies for better waste and water management
    - Educate others about the importance of sustainability